DATA 211 – Introduction to Data Science
Topic 3: Basic Statistics

Question bank: 40 multiple choice + 10 programming

Questions from DATA 211 / ISE 291 old exams and quizzes (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1.

Topic 3 MCQ (T212)

It represents categories with order.

(A) Nominal variable

(B) Ordinal variable

(C) Independent variable

(D) Dependent variable

Answer: (B). Ordinal data is categorical data whose categories have a natural order (e.g., Low, Medium, High).

My answer

Q2.

Topic 3 MCQ (T212)

Nominal variables are

(A) not affected by other variables

(B) categories with order

(C) represented using numbers for convenience

(D) dependent on other variables

Answer: (C). Nominal values (e.g., Yes = 1, No = 0) are numbers used only for convenience; they carry no mathematical meaning or order.

My answer

Q3.

Midterm 223

Suppose you are analyzing a patient's dataset that includes information on blood pressure (for example 138, 140), risk category (for example low, high), age (for example 25, 30), and blood group (for example O+, AB+). Which of the following can be the best example of nominal data?

(A) Blood pressure

(B) Risk category

(C) Blood group

(D) Age

Answer: (C). Blood group is a category with no order. Risk category is ordinal; blood pressure and age are numerical.

My answer

Q4. Quiz 1 (T253, Version A)

Which field type is best for representing data corresponding to telephone numbers?

(A) Integer

(B) Numerical

(C) Nominal

(D) Ordinal

Answer: (C). Telephone numbers are labels: adding or averaging them has no meaning, and they have no order, so they are nominal.

My answer

Q5. Mid exam v important questions

You are conducting a study and want to understand how "hours of study" affect "test scores". Which statement is true?

(A) Test score is the dependent variable and the number of "hours of study" is independent.

(B) "Hours of study" is dependent upon the independent variable "test scores".

(C) Both variables are dependent.

(D) Both variables are independent.

Answer: (A). The test score is the response (target) that depends on hours of study, which is the explanatory (input) variable.

My answer

Q6. Practice Questions Topic 3

A graph showing how many times each value occurs in a variable is called

(A) Histogram

(B) Box-plot

(C) Pie chart

(D) None

Answer: (A). A histogram shows the frequency (count) of the data falling in each value or class.

My answer

Q7.

Practice Questions Topic 3

What is the output of the following code?

```
Subj = ['M', 'P', 'M', 'P', 'P', 'P', 'P', 'P', 'M', 'M', 'P', 'M', 'P', 'M', 'M', 'C', 'C',
     'M', 'C', 'P', 'P', 'P', 'M']
import matplotlib.pyplot as plt
plt.figure(figsize=(2,2))
plt.hist(Subj)
plt.show()
```

(A) A histogram with 4 bars

(B) A histogram with 2 bars

(C) A histogram with 3 bars

(D) A pie chart with 3 sectors

Answer: (C). For categorical data the histogram has one bar per unique value: M, P and C (counts 9, 11, 3).

My answer

Q8.

[new]

A dataset has $Q_1 = 20$ and $Q_3 = 35$. Using the $1.5 \times \text{IQR}$ rule, above which value is a point an outlier?

(A) 57.5

(B) 52.5

(C) 50

(D) 42.5

Answer: (A). $\text{IQR} = 35 - 20 = 15$, so the upper fence is $Q_3 + 1.5 \times 15 = 35 + 22.5 = 57.5$.

My answer

Q9.

[new]

For the data `12, 15, 11, 18, 15, 20, 15, 14`, which statement is true?

(A) Mean = median = mode = 15

(B) Mean = 15, median = 14.5, mode = 15

(C) Mean = 15, median = 15, mode = 18

(D) The mean is greater than the median

Answer: (A). Sum = 120, so the mean is 15; sorted, the middle two values are 15 and 15, so the median is 15; 15 appears three times, so the mode is 15.

My answer

Q10.

Sample Mid

Which of the following is FALSE about a normal distribution and its histogram?

(A) All bars of the histogram are equal

(B) Distribution is symmetrical around the center

(C) Median is equal to mean

(D) There is one peak at the center

Answer: (A). A normal histogram is bell shaped: tall bars at the center and short bars in the tails, not equal bars.

My answer

Q11.

Topic 3 MCQ (T212)

The median of _____ is 4.

(A) 9, 4, 1

(B) 2, 4, 4, 6

(C) 2, 3, 5, 6

(D) All the above

Answer: (D). Sorted 1, 4, 9 has middle value 4; for 2, 4, 4, 6 the median is (4+4)/2 = 4; for 2, 3, 5, 6 it is (3+5)/2 = 4.

My answer

Q12.

Topic 3 MCQ (T212)

The mode of _____ is 4.

(A) 9, 4, 1

(B) 2, 4, 4, 6

(C) 2, 3, 5, 6

(D) All the above

Answer: (B). Only in 2, 4, 4, 6 does 4 occur more often than any other value; in the other lists every value occurs once.

My answer

Q13.

Midterm 212+213

The median of the following data is? 5, 2, 6, 1, 3, 9, 7, 1

(A) 4

(B) 1

(C) 2.5

(D) 3

Answer: (A). Sorted: 1, 1, 2, 3, 5, 6, 7, 9 ($n = 8$, even), so the median is (3+5)/2 = 4.

My answer

Q14.

Sample Mid

What is the output of the following code?

```
import numpy as np
arr = [2,4,3,1]
np.median(arr)
```

(A) 4

(B) 3

(C) 3.5

(D) 2.5

Answer: (D). Sorted: 1, 2, 3, 4, so the median is (2+3)/2 = 2.5.

My answer

Q15.

Practice Questions Topic 3

What is the output of the following code?

```
Score = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 79, 87, 87, 79, 75, 100, 82, 83, 98,
    86, 87, 71, 78, 81]
Score = np.array(Score)
print(f'The mode is {stats.mode(Score)[0]}.')
```

(A) 81, 87

(B) 81

(C) 87

(D) 100

Answer: (B). 81 and 87 both occur 3 times, but `stats.mode` returns a single value, the smallest mode: 81.

My answer

Q16.

Sample Mid

We want to find the number of cars most frequently occurring in the list using the Python `statistics` library. What will XX represent to find at least one or more values with the highest frequency?

```
from statistics import XX
Num_of_cars = [0, 1, 1, 2, 3, 3, 3, 4, 4, 5, 6, 6, 6]
most_frequent_numbers = XX(Num_of_cars)
print(f"The most frequent number(s) of cars: {most_frequent_numbers}")
```

(A) mode

(B) median

(C) mean

(D) multimode

Answer: (D). `multimode` returns all modes, here [3, 6]; `mode` returns only one value (3).

My answer

Q17.

Quiz 1, Sec 4 (TOPIC 2 & 3)

For the data

```
Score = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 79, 87, 87, 79, 75, 100, 82, 83, 98,
    86, 87, 71, 78, 81]
```

calculate the mean, median and mode (by hand).

(A) mean = 84.44, median = 82, mode = 81 and 87

(B) mean = 84.44, median = 82, mode = 81 only

(C) mean = 82, median = 84.44, mode = 81 and 87

(D) mean = 84.44, median = 83, mode = 87 only

Answer: (A). Sum = 2111, so mean = 2111/25 = 84.44; the 13th sorted value is 82; 81 and 87 each occur 3 times, so the data is bimodal (old key said 81 only, which is just what `stats.mode` prints).

My answer

Q18. Quiz 1 (T252)

Which of the following is not robust to outliers?

(A) Mean

(B) Median

(C) Interquartile Range (IQR)

(D) Mode

Answer: (A). The mean uses every value, so one extreme value moves it; the median, IQR and mode depend only on order or frequency (old key said B).

My answer

Q19. Topic 3 MCQ (T212)

Which is correct about the normal distribution?

(A) Data is distributed symmetrically around the center of all the data.

(B) The mean, median and mode are equal.

(C) It has a bell-shaped curve.

(D) All the above.

Answer: (D). The normal distribution is symmetric and bell shaped, and its mean, median and mode coincide.

My answer

Q20. Practice Questions Topic 3

A positively skewed data has a histogram with

(A) a long tail on the left

(B) equal bars

(C) symmetry about the mean value

(D) a long tail on the right

Answer: (D). In positive (right) skew a few large values stretch the right tail, and mean $>$ median $>$ mode.

My answer

Q21. Sample Mid

The standard deviation measures

(A) How close the data points are to the mean

(B) The number of times a value appears

(C) The exact difference between the highest and lowest value

(D) The direction of the data (skewness)

Answer: (A). SD is the square root of the variance: it measures spread around the mean, in the same units as the data.

My answer

Q22. Practice Questions Topic 3

The formula $\sqrt{\frac{\sum(X_i - \overline{X})^2}{N}}$ is used to compute

(A) Sample standard deviation

(B) Population standard deviation

(C) Population mean

(D) Population variance

Answer: (B). Dividing by N and taking the square root gives the population SD; the sample SD divides by $n - 1$.

My answer

Q23. Midterm 223

Suppose we have a dataset with the following values: 3, 4, 6, 7, 10. What is the sample variance of this dataset? $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$

(A) 3.5

(B) 4.05

(C) 7.5

(D) 6.5

Answer: (C). $\bar{x} = 6$; squared deviations 9, 4, 0, 1, 16 sum to 30; $s^2 = 30/4 = 7.5$.

My answer

Q24. Sample Mid

Consider a list named `height` with 50 integer values as sample data from a population. Which of the following codes will correctly provide a measure of spread of the `height` data such that the unit of the measure of spread is the same as the actual data?

(A) `np.std(height,ddof=1)`

(B) `np.var(height,ddof=1)`

(C) `np.std(height)`

(D) `np.var(height)`

Answer: (A). The SD has the same units as the data (variance is in squared units), and `ddof=1` gives the sample version.

My answer

Q25.

Quiz 1 key (Version B)

Consider the code below. Which statistic does the printed value represent?

```
import numpy as np
data = np.array([2,4,4,4,5,5,7,9])
s = np.std(data, ddof=1)
print(s**2)
```

(A) population variance

(B) sample variance

(C) sample standard deviation

(D) population standard deviation

Answer: (B). Squaring the sample SD (`ddof=1`) gives the sample variance, $32/7 \approx 4.571$ (the population variance would be 4.0).

My answer

Q26.

Quiz 1 (T253, Sec 04/05)

Two datasets have the same mean. Dataset A has a standard deviation of 8, while Dataset B has a variance of 36. Which dataset is more dispersed?

(A) Dataset A.

(B) Dataset B.

(C) Cannot be determined

(D) Both have the same dispersion

Answer: (A). The SD of B is $\sqrt{36} = 6 < 8$, so A is more spread out.

My answer

Q27.

Topic 3 MCQ (T212)

What is the output of the following code?

```
data = list(range(1,10))
x,y,z = np.percentile(data, [75 ,25, 50])
print(z)
```

(A) 1

(B) 10

(C) 5

(D) 5.5

Answer: (C). The data is 1 to 9, and z is the 50th percentile (median) = 5 (printed as 5.0); x = 7.0 is Q3 and y = 3.0 is Q1.

My answer

Q28.

Topic 3 MCQ (T212)

The upper whisker (limit) is computed as

(A) $Q_1 + 1.5\times$ Interquartile Range

(B) $Q_1 - 1.5\times$ Interquartile Range

(C) $Q_3 + 1.5\times$ Interquartile Range

(D) $Q_3 - 1.5\times$ Interquartile Range

Answer: (C). Upper limit = $Q_3 + 1.5\,\text{IQR}$; lower limit = $Q_1 - 1.5\,\text{IQR}$.

My answer

Q29.

Sample Mid

Which of the following CANNOT be correct about the whisker of a boxplot?

(A) Upper Whisker $= Q_3 + 1.5\times$ IQR

(B) Lower Whisker $> Q_1 - 1.5\times$ IQR

(C) Upper Whisker $< Q_3 + 1.5\times$ IQR

(D) Lower Whisker $< Q_1 - 1.5\times$ IQR

Answer: (D). The lower whisker stops at the smallest data point that is $\geq Q_1 - 1.5\,\text{IQR}$; it can never go below that limit.

My answer

Q30.

Practice Questions Topic 3

The following data

```
Score = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 50, 79, 87, 87, 79, 75, 100, 82, 120,
    83, 98, 86, 87, 71, 78, 81]
```

has (use `np.percentile` for the quartiles)

(A) No outlier

(B) One outlier

(C) Three outliers

(D) Two outliers

Answer: (D). $Q_1 = 78.5$, $Q_3 = 88$, IQR = 9.5, so the limits are 64.25 and 102.25; only 50 and 120 fall outside.

My answer

Q31.

Midterm 212+213

Estimate the interquartile range from the following (rotated) box plot: the whiskers end at 20 and 80, the box extends from 40 to 60, and the median line is at 50.

(A) 80

(B) 40

(C) 60

(D) 20

Answer: (D). IQR $= Q_3 - Q_1 = 60 - 40 = 20$ (the box edges); $80 - 20 = 60$ would be the whisker-to-whisker range.

My answer

Q32.

Quiz 1 (T252)

Which of the following statements about the Interquartile Range (IQR) is incorrect?

(A) IQR $= Q_3 - Q_1$.

(B) IQR is unaffected by outliers.

(C) IQR can be used to detect outliers using the $1.5\times$IQR rule.

(D) IQR is always greater than the range.

Answer: (D). The IQR covers only the middle 50% of the data, so it can never exceed the range (max − min).

My answer

Q33. Quiz 1, Sec 4 (TOPIC 2 & 3)

What is the p-value?

(A) The measure of effect size between two variables

(B) The probability of accepting the alternative hypothesis

(C) The confidence or probability value of the null hypothesis

(D) (A) and (B)

Answer: (C). As in the slides, the p-value is the confidence or probability value of the null hypothesis; a very small p-value means we reject H_0.

My answer

Q34. Topic 3 MCQ (T212)

"The sample has a normal distribution" is

(A) the null hypothesis of the U-test

(B) the null hypothesis of the t-test

(C) the null hypothesis of the Shapiro test

(D) the alternative hypothesis of the Shapiro test

Answer: (C). The Shapiro–Wilk test's H_0 is that the sample is normally distributed.

My answer

Q35. Sample Mid

Which of the following statements is NOT true about the Student's t-test?

(A) It is suitable to use with two samples when both samples do not follow the normal distribution.

(B) The null hypothesis is: the means of the two populations are not significantly different.

(C) It provides a p-value to assess the statistical significance of the observed difference in means.

(D) It is used to compare the means of two independent groups with normally distributed data.

Answer: (A). The t-test assumes normal distributions; for non-normal samples the Mann–Whitney U test is used.

My answer

Q36. Midterm 223

When conducting a Shapiro–Wilk test for sampled data, the calculated p-value was 0.03. At a significance level of 0.001, which of the following conclusions can be drawn?

(A) The sampled data does not follow a normal distribution.

(B) The sampled data has equal means.

(C) The sampled data does not have equal means.

(D) The sampled data follows a normal distribution.

Answer: (D). $p = 0.03 \geq \alpha = 0.001$, so we fail to reject H_0 (normality).

My answer

Q37.

Mid exam v important questions

In a study, you want to compare the average heights of two different plant species. You perform a Student's t-test and get a p-value of 0.15. What does this mean?

(A) There is insufficient evidence to conclude that the two species have different average heights.

(B) There is strong evidence that the two species have different average heights.

(C) You should reject the null hypothesis and conclude that the average heights of the two species are equal.

(D) The difference between the two averages is significant.

Answer: (A). $p = 0.15 \geq 0.05$, so we fail to reject H_0 (equal means): the difference is not significant.

My answer

Q38.

Sample Mid

The U-test is used to test if the means of two independent samples are significantly different when the data is not normally distributed. Which of the Python code options below returns the p-value of the U-test correctly?

(A)
```
from scipy.stats import ttest_ind as u_test
s, pval = u_test(data1, data2)
```

(B)
```
from scipy.stats import mannwhitneyu as u_test
s, pval = u_test(data1, data2)
```

(C)
```
from scipy.stats import ttest_ind as u_test
pval, s = u_test(data1, data2)
```

(D)
```
from scipy.stats import mannwhitneyu as u_test
pval, s = u_test(data1, data2)
```

Answer: (B). The U-test is `mannwhitneyu`, and it returns (statistic, p-value) in that order.

My answer

Q39.

Midterm 212+213

When using the Shapiro–Wilk test on two samples of data at the 0.05 significance level, the obtained

p-values were 0.04 for Sample_1 and 0.06 for Sample_2. Which of the following can be concluded from this result?

(A) The two samples have significantly different means.

(B) Sample_2 follows a normal distribution, but not Sample_1.

(C) Sample_1 follows a normal distribution, but not Sample_2.

(D) The two samples are independent of each other.

Answer: (B). Sample_1: $0.04 < 0.05$, so normality is rejected; Sample_2: $0.06 \geq 0.05$, so we fail to reject normality.

My answer

Q40. Quiz 1 (T253, Sec 03)

We need to compare the mean scores of two independent student groups. Preliminary analysis reveals that one of these groups does not follow a normal distribution. Which statistical test is the most appropriate in this scenario?

(A) Shapiro–Wilk Test

(B) Mann–Whitney U Test

(C) Student's t-test

(D) Paired t-test

Answer: (B). The t-test needs both samples to be normal; when one is not, use the Mann–Whitney U test.

My answer

Part B: Programming

Q41.

Set_1 (Problem B1)

Consider the following list (scores of 22 students):

```
Score_3 = [71, 70, 79, 73, 67, 68, 65, 64, 70, 50, 73, 73, 90, 52, 73, 66, 71, 60, 89, 57,
    78, 63]
```

Draw the histogram of `Score_3` using 10 bins in total. Label the axes and print the count in each bin.

Answer:

```
import matplotlib.pyplot as plt
plt.figure()
counts, edges, _ = plt.hist(Score_3, bins=10, edgecolor='black')
plt.xlabel('Score_3')
plt.ylabel('Frequency')
plt.title('Histogram of Score_3')
plt.show()
print(counts)   # [2. 1. 1. 3. 3. 8. 0. 2. 0. 2.]
print(edges)    # 50, 54, ..., 90 (bin width 4)
```

My answer

Q42.

Set_1 (Problem B3)

Calculate the mean, sample standard deviation, and range for column `Score_1`. Round to 2 decimals.

```
Score_1 = [71, 81, 79, 68, 66, 77, 73, 74, 78, 65, 79, 77, 81, 64, 77, 61, 67, 57, 97, 57,
    79, 71]
```

Answer:

```
import numpy as np
Score_1 = np.array(Score_1)
print(f"The mean is {round(Score_1.mean(), 2)}")                          # 72.68
print(f"The standard deviation is {round(np.std(Score_1, ddof=1), 2)}")  # 9.27
print(f"The range is {Score_1.max() - Score_1.min()}")                    # 40
```

My answer

Q43.

Set_1 (Problem B4)

Calculate the median, Q_1, Q_3, IQR and the whisker limits for column `Score_2`.

```
Score_2 = [71, 59, 80, 78, 68, 59, 57, 54, 62, 56, 68, 69, 79, 40, 70, 72, 75, 64, 81, 58,
    78, 56]
```

Answer:

```
import numpy as np
Q3, Q1 = np.percentile(Score_2, [75, 25])
IQR = Q3 - Q1
upperWhisker = Q3 + 1.5 * IQR
lowerWhisker = Q1 - 1.5 * IQR
print(f"Median = {np.median(Score_2)}")                    # 68.0
print(f"Q1 = {Q1}, Q3 = {Q3}, IQR = {IQR}")               # 58.25, 74.25, 16.0
print(f"Upper whisker limit = {upperWhisker}")             # 98.25
print(f"Lower whisker limit = {lowerWhisker}")             # 34.25
```

My answer

Q44.

Set_1 (Problem B5)

Consider the following data:

```
Score_1 = [71, 81, 79, 68, 66, 77, 73, 74, 78, 65, 79, 77, 81, 64, 77, 61, 67, 57, 97, 57,
    79, 71]
Score_2 = [71, 59, 80, 78, 68, 59, 57, 54, 62, 56, 68, 69, 79, 40, 70, 72, 75, 64, 81, 58,
    78, 56]
```

```
Score_3 = [71, 70, 79, 73, 67, 68, 65, 64, 70, 50, 73, 73, 90, 52, 73, 66, 71, 60, 89, 57,
    78, 63]
```

Compare `Score_1`, `Score_2` and `Score_3` by drawing box plots, all three in one figure. Label the x-axis with Score_1, Score_2 and Score_3. Then print the names of all the columns that contain one or more outliers (use the $1.5\times$IQR rule).

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
data = {'Score_1': Score_1, 'Score_2': Score_2, 'Score_3': Score_3}
fig = plt.figure(figsize=(9, 6))
ax = fig.add_subplot(111)
ax.boxplot(list(data.values()))
ax.set_xticklabels(list(data.keys()))
plt.title('Scores')
plt.show()

for name, x in data.items():
    Q3, Q1 = np.percentile(x, [75, 25])
    IQR = Q3 - Q1
    out = [v for v in x if v > Q3 + 1.5*IQR or v < Q1 - 1.5*IQR]
    if out:
        print(name, 'has outliers:', out)
# Output: Score_3 has outliers: [50, 90, 89]
```

My answer

Q45.

Midterm-213 (Problem B-1)

Given the following samples of scores in two exams:

```
final_scores   = [45, 60, 70, 70.5, 60.5, 60, 61, 70, 59, 80, 81, 71, 50, 60, 55, 63]
midterm_scores = [60, 80, 60, 70.5, 60.5, 70, 61, 72, 77, 81, 79, 87, 70, 53, 40, 50]
```

Draw the box plots for the two samples (in one figure), then identify the upper and lower whisker limits for each sample.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111)
ax.boxplot([final_scores, midterm_scores])
```

```
ax.set_xticklabels(['final_scores', 'midterm_scores'])
plt.title('Box Plot')
plt.show()

for name, x in [('final', final_scores), ('midterm', midterm_scores)]:
    Q3, Q1 = np.percentile(x, [75, 25])
    IQR = Q3 - Q1
    print(f"{name}: upper = {Q3 + 1.5*IQR}, lower = {Q1 - 1.5*IQR}")
# final:   upper = 85.6875, lower = 44.1875
# midterm: upper = 103.75,  lower = 33.75
```

My answer

Q46. Midterm-213 (Problems B-2, B-3)

Given the following samples of scores in two exams:

```
final_scores   = [45, 60, 70, 70.5, 60.5, 60, 61, 70, 59, 80, 81, 71, 50, 60, 55, 63]
midterm_scores = [60, 80, 60, 70.5, 60.5, 70, 61, 72, 77, 81, 79, 87, 70, 53, 40, 50]
```

(a) Is it reliable to use the t-test to test if the means of the two scores are significantly different? Support your answer using the appropriate statistical test. (b) Following your conclusion in (a), use the appropriate test to show whether the performance of the students in the two exams is significantly different. State the null hypothesis and comment on the result ($\alpha = 0.05$).

Answer:

```
from scipy.stats import shapiro, ttest_ind

# (a) Shapiro-Wilk, H0: the sample is normally distributed
for name, x in [('final', final_scores), ('midterm', midterm_scores)]:
    _, pval = shapiro(x)
    print(name, 'p-value', pval)          # 0.51 and 0.73
    if pval < 0.05:
        print('  Reject H0: not normal')
    else:
        print('  Fail to reject H0: normal')
# Both samples are normal, so the t-test is reliable.

# (b) Student's t-test, H0: the means of the two samples are equal
_, pval = ttest_ind(final_scores, midterm_scores, equal_var=False)
print('p-value', pval)                    # 0.40
if pval < 0.05:
    print('Reject H0: performance is significantly different')
else:
    print('Fail to reject H0: no significant difference in performance')
```

My answer

Q47. Quiz-1 sample, Part 2 (TOPIC 2 & 3)

Given the following lists:

```
quiz_1 = [8.5, 6.4, 7.2, 7.3, 6.5, 6.0, 6.1, 7.2, 6.6, 8.0, 8.9, 7.7, 5.0, 6.3]
quiz_2 = [6.3, 8.4, 6.2, 7.4, 6.5, 7.0, 6.1, 7.2, 7.7, 8.1, 7.9, 8.7, 7.0, 5.3]
```

Do the following using Python: (1) Find the median score of each quiz. (2) Under the t-test: (a) state the null hypothesis for comparing the two quizzes; (b) test your hypothesis and comment on the result. (3) Support your conclusion in (2) visually, using a boxplot.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind

# (1)
print('Median quiz_1 =', np.median(quiz_1))   # 6.9
print('Median quiz_2 =', np.median(quiz_2))   # 7.1

# (2a) H0: the mean of quiz_1 equals the mean of quiz_2 (normal data assumed)
# (2b)
_, pval = ttest_ind(quiz_1, quiz_2)
print('p-value', pval)                         # about 0.70
print('Reject H0' if pval < 0.05 else 'Fail to reject H0')   # Fail to reject H0

# (3) The boxes overlap heavily, so the two quizzes look similar
fig = plt.figure(figsize=(5, 6))
ax = fig.add_subplot(111)
ax.boxplot([quiz_1, quiz_2])
ax.set_xticklabels(['quiz_1', 'quiz_2'])
plt.show()
```

My answer

Q48.

Quiz #01 (Problem B)

Run this code:

```
Math_Score   = [56, 83, 55, 45, 100, 35, 95, 80, 60, 80]
ISE291_Score = [70, 55, 100, 40, 100, 68, 85, 65, 74, 53]
```

(B.1) Draw the box plots of `Math_Score` and `ISE291_Score`.
(B.2) Check if `Math_Score` and `ISE291_Score` follow the normal distribution (reject the null hypothesis for a p-value less than 5%). Print the p-value and the null hypothesis result.

Answer:

```python
import matplotlib.pyplot as plt
from scipy.stats import shapiro

# B.1
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111)
ax.boxplot([Math_Score, ISE291_Score])
ax.set_xticklabels(['Math_Score', 'ISE291_Score'])
plt.title('Box Plot')
plt.show()

# B.2  H0: the sample follows a normal distribution
for name, x in [('Math', Math_Score), ('ISE 291', ISE291_Score)]:
    _, pval = shapiro(x)
    print(name, 'p-value', pval)       # 0.643 and 0.689
    if pval < 0.05:
        print('  We reject the null hypothesis (not normal)')
    else:
        print('  We fail to reject the null hypothesis (normal)')
```

My answer

Q49. Quiz 1 (Problem 2.2)

The following data is about 12 patients on whom 3 different types of sleeping pills (A, B and C) are tested. The time to sleep is reported for each patient for each pill.

```
Time_A = [13, 12, 13, 11, 12, 13, 14, 14, 11, 12, 15, 10]
Time_B = [12, 13, 16, 17, 12, 14, 13, 15, 16, 17, 16, 17]
Time_C = [20, 24, 20, 22, 21, 21, 25, 20, 23, 25, 22, 24]
```

Show the boxplot of the time to sleep for each sleeping pill in a single graphical layout. Print the median and IQR of each pill and interpret your findings.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111)
ax.boxplot([Time_A, Time_B, Time_C])
ax.set_xticklabels(['A', 'B', 'C'])
ax.set_ylabel('Time to sleep')
plt.title('Sleeping Pills')
plt.show()

for name, x in [('A', Time_A), ('B', Time_B), ('C', Time_C)]:
    Q1, Q2, Q3 = np.percentile(x, [25, 50, 75])
    print(name, 'median', Q2, 'IQR', Q3 - Q1)
# A: median 12.5, IQR 1.5 | B: median 15.5, IQR 3.25 | C: median 22.0, IQR 3.25
# Interpretation: no pill has outliers. Pill A has the lowest median time
# and the smallest spread, so it is the most effective and consistent;
# pill C takes the longest time to sleep.
```

My answer

Q50. [new]

Write a function `compare_groups(a, b, alpha=0.05)` that first runs the Shapiro–Wilk test on both samples. If both are normal, it compares them with Student's t-test; otherwise it uses the Mann–Whitney U test. The function returns the name of the test used, its p-value, and the decision ("reject H0" or "fail to reject H0"). Apply it to:

```
group_A = [12, 15, 14, 13, 16, 15, 14, 13, 15, 14]
group_B = [10, 11, 10, 12, 11, 10, 35, 40, 11, 10]
```

Answer:

```
from scipy.stats import shapiro, ttest_ind, mannwhitneyu

def compare_groups(a, b, alpha=0.05):
    p_a = shapiro(a)[1]
    p_b = shapiro(b)[1]
    if p_a >= alpha and p_b >= alpha:
        test = "Student's t-test"
        pval = ttest_ind(a, b)[1]
    else:
        test = 'Mann-Whitney U test'
        pval = mannwhitneyu(a, b)[1]
    decision = 'reject H0' if pval < alpha else 'fail to reject H0'
    return test, pval, decision

print(compare_groups(group_A, group_B))
# group_B is not normal (Shapiro p ~ 0.00003), so the U test is used:
# ('Mann-Whitney U test', 0.027, 'reject H0') -> the distributions differ
```

My answer